

Benjamin Schmidt

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EDUCATION

University of California, Berkeley

Bachelor of Science in Mechanical Engineering, Minor in Electrical Engineering and CS

Berkeley, CA

Class of 2029

SKILLS

Tools: Solidworks, Fusion, ANSYS Fluent, MATLAB, Excel

Software: Python, C++, SQL, Java, ROS, PX4, ArduPilot, TensorFlow, Pytorch

Hardware: High Frequency RF, Raspberry Pi

EXPERIENCE

Research Assistant

August 2024 – April 2026

Degrendele Lab

Fremont, CA

- Developed Physics Informed Neural Networks (PINNs) and Physics Informed Neural Operators (PINOs) for Aerodynamic Shape Optimization under NASA researcher
- Created optimization algorithms for shape modification, showing 14x improvement in aerodynamic properties
- Increased optimization speed by 113x compared to traditional CFD methods whilst maintaining high accuracy
- 2nd author of research [paper](#) published at the MIT Undergraduate Research Technology Conference

Mechanical Sub-team Member

Jan 2025 – April 2026

Longhorn Robotics

San Jose, CA

- Used CAD software to design and prototype robot chassis, drivetrain, and intake system for school's First Tech Challenge robotics team
- Simulated and constructed parts for the robot and ran hundreds of tests to ensure accuracy and structural integrity of the system
- Assisted in writing software for and testing micro-controller based embedded system onboard

Design Specialist

May 2025 – July 2025

Dinobikes

San Jose, CA

- Designed prototype glasses for use in upcoming Augmented Reality sports product
- Performed Finite Element Analysis, optimized design for weight and structural integrity
- Constructed mounting basis for electronics and AR optical projector

PROJECTS

EnviDrones | *Solidworks, Python, Raspberry Pi, ROS, ArduPilot, TensorFlow*

Jan 2025 – Jul 2026

- Created research and development team focused on developing high performance autonomous drones for environmental surveying
- Constructed fully autonomous UAV systems, using stereo based navigation to map environment and move around obstacles and in GPS denied regions
- Trained and tested machine learning based tree density algorithm, demonstrating 85% accuracy in tested environments and improved detection compared to baseline models
- Presented work to faculty at UC Berkeley, ran tests and demo in local parks

RoboDefect | *Python, Numpy, Scikit*

Jun 2025 – Jul 2025

- Constructed and ran a comparative analysis on machine learning models and methods for robotic arm failure detection under guidance of Stanford professor
- Demonstrated effectiveness of Tuned Random Forest algorithm in comparison to various traditional techniques for predicting failure based on real-time data
- Compiled test results using UC Irvine dataset and presented analysis in a research poster

AttendFlow | *High Frequency RF, Flask, Javascript, SQL*

Jun 2025 – Jul 2025

- Built and configured Ultra-High Frequency RFID scanning system to read student IDs remotely and store attendance records
- Developed full-stack web application for school attendance based on the 5-Star App and an API to communicate with the card reader
- Created features for event management, schedule adjustment, and student information handling.
- Presented project to school district and discussed methods to implement at a wide scale